E-COMMERCE RETURN RATE REDUCTION ANALYSIS

Business intelligence, customer-return analysis and product-level return-risk modelling

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1. Introduction

Project: Data Analytics + Machine Learning

Product returns affect revenue, logistics cost, inventory planning and customer experience. This project analyses order and return behaviour across product categories, marketing channels, delivery performance, customer ratings, regions and suppliers. It combines descriptive analytics with a Logistic Regression model that generates product-level return-risk scores for investigation.

2. Abstract

The project uses a cleaned order-level dataset of 100,000 records. A total of 14,047 orders were returned, producing an overall return rate of 14.05%. Exploratory analysis identifies segments with comparatively higher observed return rates. A Logistic Regression classifier was evaluated on a held-out test set, and scores were aggregated into a product-level risk table covering 2,500 products.

3. Tools Used

Area	Tools
Data preparation	Python, Pandas, NumPy
Visualisation	Power BI, Matplotlib/ Seaborn
Database querying	SQL
Machine learning	Scikit-learn, Logistic Regression
Reporting	DAX, Power BI dashboard, PDF report

4. Steps Involved in Building the Project

- **Data understanding:** reviewed order-level fields, target variable and product-risk output.
- **Data cleaning:** removed 70 duplicate records from 100,070 raw rows, resulting in 100,000 records; checked duplicate Order_IDs and key missing fields.
- **Exploratory analysis:** calculated return rates by category, marketing channel, delivery group, rating group, region, state and supplier.
- **Feature preparation:** excluded post-return fields such as Return_Reason and Return_Days_After_Delivery to reduce target leakage.
- **Model development:** compared Logistic Regression with tree-based models using cross-validation and evaluated the selected model on a held-out test set.
- **Product scoring:** aggregated order-level scores into a product-level table and flagged the top quartile as high-risk candidates.
- **Power BI reporting:** created performance, customer/fulfilment, geography, supplier/product, risk, drill-through and recommendation views.

5. Key Analytical Findings

Area	Observed result	Business interpretation
Overall	14,047/100,000; 14.05%	Overall return level in the cleaned dataset.
Category	Fashion: 14.99%	Highest observed category rate among listed categories.
Marketing	Social Media: 15.34%	Highest observed channel rate among listed channels.
Delivery	Over 6 days: 16.91%	Higher than upto--3-days group at 11.88%.
Rating	Low rating (1-3): 15.00%	Higher than medium and high rating groups.
Product risk	2,500 scored; 625 top-quartile candidates	Prioritisation shortlist, not a >50% probability claim.

6. Machine Learning Model Evaluation

Selected model: Logistic Regression. Test-set results at threshold 0.45: PR-AUC 0.1677, ROC-AUC 0.5559, precision 14.49%, recall 87.61% and F1-score 0.2486. Positive-class prevalence is approximately 14.05%, so PR-AUC is only modestly above the prevalence baseline. The threshold favours recall and creates many false positives. Treat this as an exploratory screening model, not a production-ready decision system.

7. Power BI Dashboard - Page Headings

Page	Dashboard page title
01	Executive Overview - E-commerce Return Performance
02	Customer Experience & Fulfilment Analysis
03	Geographic Return Performance & Market Insights
04	Supplier & Product Quality Intelligence
05	Predictive Return Risk & Model Evaluation
06	Product-Level Return Investigation
07	Return Drivers & Category Trend Analysis
08	Business Recommendations & Model Governance

8. Conclusion & Recommendations

Fashion, Social Media orders, longer delivery times and low customer ratings are segments for further investigation. Teams can review product descriptions and sizing, campaign targeting, delayed shipments and supplier/product quality. These are observed associations, not proof of causation.

Model governance: validate product-key uniqueness and category mapping; assess probability calibration with a calibration curve and Brier score; monitor precision/recall at a business-approved threshold; and test performance over time before operational deployment.